Yuri Bendall

Materials Engineering | Mechatronics & Hardware Systems | Simulation & R&D
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UBC Materials Engineering student focused on full-systems design, where materials science drives both mechanical and electrical performance. Experienced across the complete development cycle, from Ansys CFD/FEA simulation, microstructural characterization, and custom PCB fabrication to avionics integration and 500+ unit commercial quality validation.

EXPERIENCE

Founder & Product Lead | Solestia Inc. (Portable Solar Hardware Venture) **Aug 2024 - Present**

- Directed functional testing, root-cause failure analysis, and regulatory compliance checks across 500+ production units, identifying assembly defects and authoring Standard Operating Procedures (SOPs) to improve hardware reliability.
- Managed overseas contract manufacturer sourcing, component specifications, packaging compliance, and international freight logistics. Built the company's web infrastructure and produced commercial video assets, translating complex technical specifications into accessible customer-facing content.

Avionics Power & Controls Lead | UBC AeroDesign **Sept 2023 - Oct 2024**

- Led a 6-person subteam responsible for multi-layer PCB development and designing, wiring, and integrating the propulsion, high-current power distribution, RF communication, and flight-control subsystems for the micro-class aircraft.
- Conducted wind tunnel propulsion tests and rotating-domain Ansys CFD/FEA propeller simulations to evaluate thrust efficiency, blade root stress, vortex formation, and flow separation.
- Diagnosed and resolved electrical and control-surface faults under tight flight-line timelines using multimeters and systematic debugging, while enforcing shop safety and preventative maintenance.
- Managed micro-class subteam operations by organizing regular syncs, overseeing hardware inventory, and coordinating closely with structural and pilot teams to ensure seamless aircraft integration.

Produce Clerk & Shift Trainer | Old Country Market **Mar 2018 - Jul 2023**

- Handled daily inventory rotation, pallet delivery sign-offs (50+ lb manual handling), independent store opening and closing procedures, and point-of-sale reconciliation.
- Trained incoming staff on workplace safety SOPs, equipment handling, and sanitation standards.

ENGINEERING PROJECTS

Custom PCB Design & Aqueous Chemical Etching **2026**

- Designed custom circuit schematics and single-layer PCB layouts in KiCad (integrating a Raspberry Pi Pico) and Altium Designer, optimizing footprint geometry, power delivery, and trace routing for manufacturability.
- Etched copper-clad boards using hydrochloric acid and hydrogen peroxide aqueous solutions in home and supervised UBC Hydrometallurgy Lab settings, controlling reaction kinetics and temperature while safely neutralizing hazardous copper byproducts.
- Completed THT soldering, visual defect inspection, and multimeter debugging to verify board functionality.

Materials Characterization & Morphology of Lapis Lazuli | UBC MTRL 451 **2026**

- Isolated lazurite, calcite, and pyrite phases by combining X-ray diffraction (XRD) peak indexing with Scanning Electron Microscopy (SEM) and optical microscopy.
- Paired Secondary Electron (SE) fracture topography with Backscattered Electron (BSE) atomic-number contrast to correlate brittle cleavage mechanisms and differential hardness relief across polished cross-sections.

Co-Project Manager - Polymer Barrier Coating R&D | UBC MTRL 320 **2026**

- Co-leading project scheduling and technical formulation targeting low-surface-energy polymer wettability, viscosity control, and surface adhesion.

Machine Learning & Real-Time Computer Vision **2026**

- Trained and deployed a custom Convolutional Neural Network (CNN) in Python using TensorFlow Lite, MediaPipe, and OpenCV on a self-curated webcam dataset to map live hand gestures to system-level keystrokes, packaged via PowerShell.

Vertical Axis Wind Turbine (VAWT) **2024**

- Modeled a Vertical Axis Wind Turbine assembly in SolidWorks and ran Ansys CFD simulations to analyze aerodynamic blade performance and wake flow behavior.
- Performed material-specific Ansys Finite Element Analysis (FEA) to verify structural stress margins under operational loads before 3D-printing modular components and completing physical assembly.

Robotics & Mechatronic Actuation Prototypes (RC Drone & Rover) **2024**

- Designed and 3D-printed custom RC drone and rover chassis; integrated Arduino and Raspberry Pi controllers across UART, SPI, and I2C sensor buses.
- Implemented wireless RX/TX communication links and diagnosed real-world RF interference, latency, and signal-integrity constraints affecting closed-loop control.

EDUCATION

University of British Columbia | B.A.Sc in Materials Engineering | Expected May 2029

- **Awards**: UBC Centennial Leaders Entrance Scholarship (awarded for academic achievement, technical initiative, and community leadership).
- **Selected Coursework**: Microstructural Analysis Laboratory (MTRL 451), Economic Aspects of Materials Engineering (MTRL 455), Materials Design & Project Management (MTRL 320), Welding & Joining Processes (MTRL 472)

TECHNICAL SKILLS & TOOLS

- **Materials & Wet Lab**: X-Ray Diffraction (XRD), Scanning Electron Microscopy (SE/BSE), Optical Microscopy, Metallographic Sample Prep (Polishing/Sectioning), Uniaxial Tensile & Hardness Testing, Wet Chemical Etching (HCl/H2O2), WHMIS & Hazardous Byproduct Neutralization.
- **Electronics & Mechatronics**: Altium Designer, KiCad, Schematic Capture & Multi-Layer PCB Layout, Microscopic SMT & THT Soldering/Rework, Oscilloscopes, Digital Multimeters, Bench Power Supplies, Electronic Loads, Raspberry Pi, Pico, Arduino, UART / SPI / I2C, RF Telemetry, High-Current LiPo Wire Harnessing.
- **Mechanical, CAD & Simulation**: SolidWorks, Autodesk Fusion, Onshape, FreeCAD, Ansys (Fluent CFD & Structural FEA), Abaqus, Simufact Welding, LTspice, Wind Tunnel Testing, FDM 3D Printing (Bambu Lab, Ultimaker Cura), Laser Cutting, Composite & Balsa Layup, Drill Press, Band Saw, Milling & Welding Equipment.
- **Software & Operations**: Python (TensorFlow, PyTorch, OpenCV, MediaPipe, LLM/RAG APIs, NumPy/Pandas), C++, MATLAB, JavaScript, HTML/CSS, PowerShell, Git, Jupyter Notebooks, Excel Inventory Modeling, SOPs